Analysis of A/B Test Results

E-Commerce Conversion Rate Experiment

Data Scientist: A/B Test Analysis Team

69,889

Total Users

13.0%

Overall Conv. Rate

3

Countries

Agenda

01

Experiment Setup

Dataset overview, participant counts, and geographic distribution of users

02

Conversion Rates

Probability analysis by treatment group and country, including interaction effects

03

Hypothesis Testing

Simulation-based p-value, statistical significance, and business recommendation

04

Expanded EDA & Logistic Regression

Propensity analysis and confounding variable investigation

How Was the Experiment Implemented?

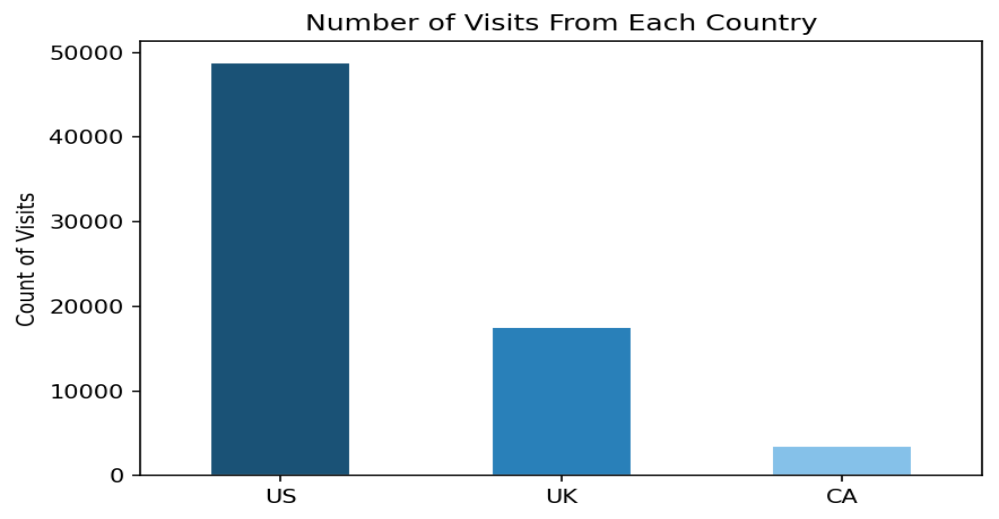
35,211
Treatment (Variant) Visitors

34,678
Control Visitors

69,889
Total Participants

0
Missing Values

Geographic Distribution of Visitors



Dataset Summary

Columns:	country, group, converted
Non-categorical column:	converted (int64: 0 or 1)
US visitors:	48,850 (69.9%)
UK visitors:	17,551 (25.1%)
CA visitors:	3,488 (5.0%)
Groups are balanced:	~50% treatment, ~50% control

Conversion Rates by Country & Group

	US U.S.	GB U.K.	CA Canada
Control	10.7%	10.2%	9.4%
Treatment	**15.8%**	**14.9%**	**15.4%**

Treatment Lift vs. Control:

U.S.

+5.1 pp (+47.7%)

U.K.

+4.7 pp (+45.9%)

Canada

+6.0 pp (+63.1%)

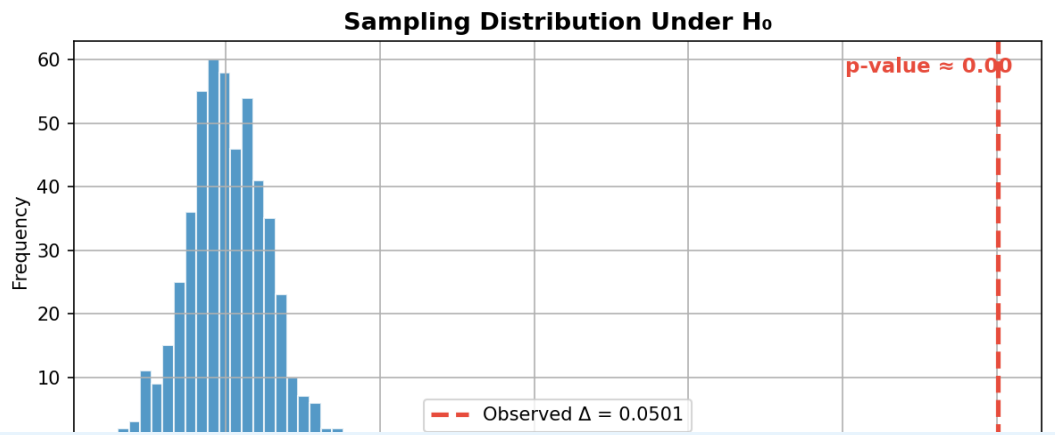
Executive Summary

The treatment page consistently outperforms the control page in every country. The lift ranges from ~5–6 percentage points across all regions, suggesting treatment is effective regardless of geography. While the US has slightly higher base conversion rates, Canada shows the largest relative improvement. No significant interaction between country and treatment group was observed — the treatment benefit appears uniform across markets.

Experiment Results

15.5%	10.5%	+5.0 pp	< 0.001
Treatment Conversion Rate	Control Conversion Rate	Delta (Treatment vs. Control)	p-value ($\alpha = 0.05$)

Sampling Distribution Under H_0



H_0: p_treatment − p_control ≤ 0 | H_1: p_treatment − p_control > 0

Conclusion

✅ We REJECT H_0 at $\alpha = 0.05$

📈 Treatment conversion (15.5%) is significantly higher than control (10.5%)

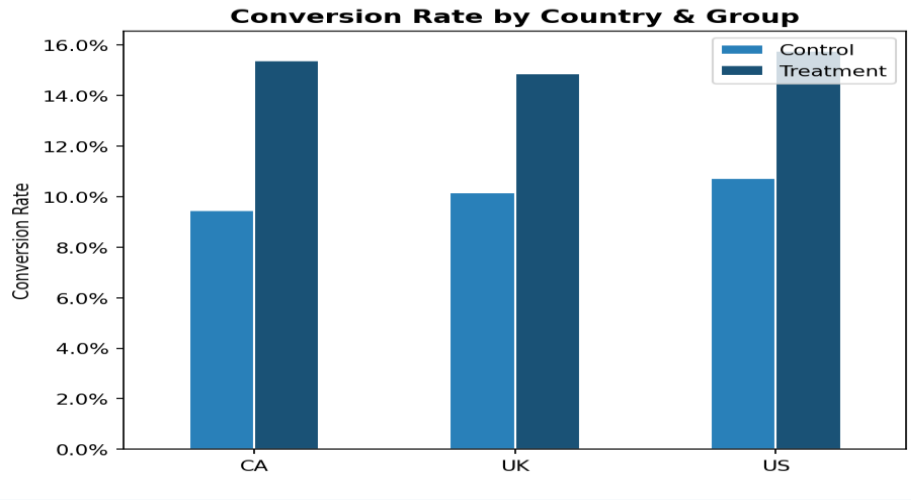
📊 p-value ≈ 0.000 — extremely strong statistical evidence

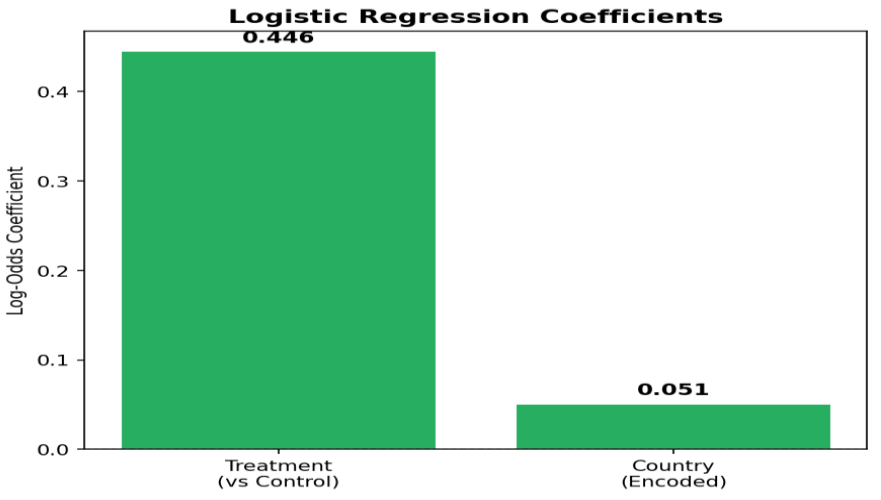
🌍 Effect is consistent across all three countries

💡 Recommendation: IMPLEMENT the new treatment page — it meaningfully increases conversions

Expanded EDA & Logistic Regression

Conversion Lift by Country & Logistic Regression Coefficients





Logistic Regression Findings

A logistic regression model was fitted using treatment group and country (encoded) as predictors of conversion. The treatment coefficient (log-odds = +0.446) is statistically meaningful and positive — confirming the treatment page significantly increases the probability of conversion. The country coefficient (+0.051) is very small, indicating country alone has minimal effect on conversion rates. This supports our hypothesis testing conclusion: the treatment effect is real and not explained by country as a confounding variable. Model accuracy: 87.0%.